

Jacob Kurtzman

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SUMMARY

Senior Data Engineer with **11** years of experience designing **cloud-native data platforms**, building **scalable ETL/ELT pipelines**, and modernizing analytics infrastructure across **AWS** and **Azure**. Proven ability to architect **Lakehouse solutions**, implement **dimensional data models**, and lead **data governance** and **quality frameworks** in regulated industries. Adept at translating business needs into robust data systems that power **BI**, **ML**, and **real-time decision-making**. Known for mentoring teams, optimizing performance, and aligning **data strategy** with enterprise goals.

TECHNICAL SKILLS

- **Programming Languages:** Python, Go, JavaScript/TypeScript, Bash/Shell
- **Backend Frameworks & APIs:** FastAPI, Gin, gRPC/REST
- **Cloud & Infrastructure:** AWS (ECS, Lambda, S3), Azure, GCP, Kubernetes, Docker
- **Security & Auth:** OAuth2, OIDC, JWT
- **CI/CD & DevOps:** GitHub Actions, Terraform, Infrastructure-as-Code, pipelines, test automation
- **Observability & Diagnostics:** Prometheus, Grafana, ELK
- **Architecture Patterns:** Microservices (DDD, service contracts), Event-Driven (Kafka, SQS, SNS)
- **Platform Engineering:** Internal tooling, golden paths
- **Data Warehousing:** Dimensional Modeling (Star Schema, Snowflake Schema), Data Lakes

WORK EXPERIENCE

Senior Data Engineer | TheoremOne

San Francisco, CA | 09/2023 – Present

- Designed and implemented a scalable **Big Data Lakehouse architecture** based on **Medallion principles** across **AWS**, enabling unified ingestion, transformation, and analytics for structured and semi-structured data reducing data duplication by 40%.
- Architected cross-cloud data pipelines leveraging **GCP BigQuery**, **Cloud Storage**, and **Pub/Sub**, enabling real-time ingestion and analytics across multi-tenant environments with unified governance and cost controls.
- Architected and modernized enterprise data platforms by transitioning from fragmented reporting systems to **dimensional-modeled data warehouses**, improving data accessibility, consistency, and query performance.
- Led development of **interactive Power BI dashboards** integrated with **Redshift** and **Google BigQuery**, transforming static reports into automated, real-time analytics tools for executive decision-making.

- Collaborated with product owners, analysts, and engineering leads to translate business requirements into **scalable data architecture**, technical specifications, and delivery roadmaps.
- Developed and enforced **data governance frameworks**, including metadata management, lineage tracking, and access controls using **AWS Lake Formation**.
- Aligned data architecture strategies with evolving business goals, ensuring compliance with **HIPAA**, **SOC 2**, and internal audit standards while optimizing for performance and cost.
- Engineered scalable **Delta Lake pipelines** in **Databricks** using **PySpark** and **SQL**, enabling ACID-compliant ingestion, transformation, and analytics across structured and semi-structured datasets.
- Designed and implemented a **business-aligned Data Quality (DQ) framework**, integrating rule engines and anomaly detection to automate validation and build trust in enterprise datasets.
- Migrated legacy analytics systems to **modern cloud-native data warehouses** (Snowflake, Redshift, Synapse), refactoring **ETL logic** and **dimensional models** for performance and scalability.
- Partnered with cross-functional teams to define **ETL/ELT pipelines** using **AWS Glue**, **dbt**, and **Airflow**, enabling seamless data movement across OLTP systems, lakes, and warehouses.
- Led migration of **legacy ETL jobs** to **GCP-native services**, refactoring **Python** and **SQL logic** for BigQuery optimization, reducing query latency by 45% and improving pipeline reliability.
- Built reusable pipeline templates and modular transformation layers, accelerating onboarding and reducing duplication across engineering teams.
- Led performance optimization initiatives in **Databricks** notebooks and jobs, tuning cluster configurations, caching strategies, and query plans to support high-concurrency workloads.
- Designed and delivered **training programs**, **best practice audits**, and **knowledge transfer sessions**, fostering a culture of continuous learning and technical excellence.
- Implemented **CI/CD pipelines** for data workflows using **GitHub Actions** and **Terraform**, enabling automated testing, deployment, and rollback for data infrastructure.
- Integrated **S3**, **Redshift**, and **DynamoDB** with **custom Python SDKs** to standardize access patterns and improve developer productivity.
- Led performance tuning initiatives across **Redshift** and **Snowflake**, optimizing sort keys, distribution styles, and query plans to support high-concurrency analytics.
- Built observability dashboards using **CloudWatch**, **Grafana**, and **QuickSight**, tracking pipeline health, SLA adherence, and data freshness across environments.
- Conducted architecture reviews and cost optimization workshops, reducing cloud spend by 30% while maintaining performance benchmarks.
- Mentored junior and mid-level engineers on **data modeling**, **cloud architecture**, and **governance**, driving team velocity and platform maturity.
- Piloted use of **LLM-based data documentation tools** to auto-generate lineage, schema summaries, and pipeline annotations from code and metadata.
- Supported executive reporting and strategic planning by delivering curated datasets and analytics layers for finance, operations, and product teams.

Data Engineer | UnitedHealth Group

Minnetonka, MN | 10/2018 – 08/2023

- Designed and maintained scalable **ETL pipelines** using **AWS Glue**, **Python**, and **Spark**, ingesting multi-terabyte datasets from claims, provider, and eligibility systems into Redshift and S3-based data lakes.
- Engineered real-time ingestion workflows using **Kinesis Data Streams**, **Firehose**, and **Lambda**, enabling sub-second analytics for care coordination and fraud detection.
- Modeled healthcare data using **dimensional schemas** and **partitioned tables** in **Redshift** and **Snowflake**, supporting actuarial forecasting, clinical dashboards, and regulatory reporting.

- Developed **PySpark jobs** on **EMR** to process encounter and claims data, implementing partition pruning and broadcast joins to reduce batch latency by 60%.
- Automated data quality checks with **Lambda**, **CloudWatch Events**, and **SNS alerts**, flagging anomalies in provider feeds and triggering remediation workflows.
- Orchestrated complex workflows using **Apache Airflow** and **AWS Step Functions**, integrating cross-region data movement, transformation, and SLA monitoring.
- Implemented **S3 lifecycle policies**, **encryption (KMS)**, and **bucket-level access controls** to ensure HIPAA compliance and secure PHI storage.
- Built internal Python libraries and wrappers for standardized access to **DynamoDB**, **S3**, **Redshift**, and **Glue Catalog**, improving developer velocity and reducing onboarding time.
- Design and Develop **Azure Data Factory / Pipelines**, uploading in **Azure Data Lake**, Data analyses in **DataBricks**.
- Led migration of legacy SSIS and Informatica pipelines to **AWS-native ETL**, reducing infrastructure costs by 40% and improving deployment agility.
- Integrated **Athena** and **Presto** for ad-hoc querying of raw claims and eligibility data, enabling actuarial teams to iterate faster on risk models.
- Partnered with data scientists to deliver feature-ready datasets for predictive modeling using **SageMaker**, **Feature Store**, and **custom preprocessing pipelines**.
- Created **CloudFormation** and **Terraform templates** to provision data lake infrastructure with audit-compliant tagging, encryption, and IAM boundaries.
- Tuned Redshift queries using **sort keys**, **distribution styles**, and **WLM queues**, improving dashboard responsiveness and concurrency under load.
- Implemented **Glue Data Catalog** and **Lake Formation** for metadata management, access control, and lineage tracking across 100+ datasets.
- Built operational dashboards in **QuickSight** and **Grafana**, visualizing pipeline health, data freshness, SLA compliance, and anomaly trends.
- Collaborated with security and compliance teams to enforce **least-privilege IAM roles**, **resource policies**, and **audit logging** across all data assets.
- Supported migration of on-prem Oracle workloads to **Aurora PostgreSQL**, refactoring stored procedures and optimizing schema for cloud-native performance.
- Documented data lineage, transformation logic, and pipeline dependencies using **Confluence**, **Swagger**, and **GitHub Wikis**, improving cross-team transparency and audit readiness.
- Mentored junior engineers on **AWS best practices**, **data modeling**, **observability**, and **CI/CD**, fostering a culture of technical excellence and ownership.
- Presented quarterly data platform reviews to leadership, mapping technical improvements to cost savings, operational KPIs, and compliance metrics.

Junior Data Engineer | Dollar General

Goodlettsville, TN | 07/2013 – 09/2018

- Assisted in designing and developing **ETL workflows** using **Talend** to ingest structured and semi-structured data into the enterprise data warehouse.
- Created and optimized **PL/SQL procedures** and **complex queries** to improve data retrieval performance and reduce processing latency.
- Supported development of **Spark jobs** in **Scala** and **Python** for batch processing on **AWS EMR**, handling millions of records across retail and logistics domains.
- Built interactive **Tableau dashboards** to visualize sales trends, inventory metrics, and operational KPIs for business stakeholders.

- Automated file ingestion and transformation using **Talend FTP components**, improving reliability and reducing manual intervention.
- Participated in building **Azure Data Factory pipelines** and integrated with **Azure Data Lake Storage** for scalable data storage and retrieval.
- Collaborated with senior engineers on **data cleansing, transformation, and integration** tasks to ensure consistency and accuracy across systems.
- Maintained and scheduled **SSIS packages** and **SQL Server Agent jobs**, automating nightly data loads and reducing manual overhead.
- Helped design **dimensional data models** (star/snowflake schemas) to support efficient reporting and analytics across finance and operations.
- Documented **ETL** logic, transformation rules, and pipeline dependencies to support onboarding and cross-team collaboration.
- Developed **parameterized Talend jobs** for **reusable ingestion patterns**, reducing onboarding time for new data sources and improving **maintainability** across **retail data domains**.
- Performed **root cause analysis** on **data discrepancies** across staging and reporting layers, collaborating with **QA** and **business analysts** to ensure **data integrity** and **trust**.
- Assisted in migrating **legacy Excel-based reports** to **automated Tableau dashboards**, improving refresh frequency from weekly to daily and enhancing stakeholder visibility.
- Contributed to **performance tuning** of **SQL Server queries** and **indexes**, reducing report generation time by 30% for high-volume transactional datasets.
- Monitored pipeline health using **AWS CloudWatch**, resolving data load failures and escalating issues to senior engineers.
- Communicated regularly with project managers and team leads to report progress, surface blockers, and align on delivery milestones.

EDUCATION

University of Kansas, Lawrence, KS | 09/2009 – 06/2013

Bachelor of Science